
Advisor: Bernardo Guimarães

Conditional Cash Transfers as Stabilization Tool: a HANK Approach

1. Model

The setup is a standard New Keynesian model with incomplete markets and labor supply risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms join in a formal-informal setup, and the rest of the economy follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady-state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity and discount factor.

Households enjoy a separable CRRA utility over consumption $c_{i,t}$ and worked hours $h_{i,t}$, which we define $u(c_{i,t}, v_{i,t}) = u(c_{i,t}) - v(h_{i,t})$ as

$$\max_{c_{i,t}, h_{i,t}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{c_{i,t}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} - \psi \frac{h_{i,t}^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}} \right],$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

A household faces idiosyncratic labor productivity shocks $e_{i,t}$ with $\mathbb{E}[e_{i,t}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, and we fix ω_I impatient households. They can work in the formal or informal sector, or be unemployed $s \in \{F, I, U\}$. Each sector has a specific productivity $\theta_{i,t}^s$ which their wage depends on, re-drawn on each new offer, with $\mathbb{E}[\theta_{i,t}^F] = 1$.

Formal workers earn labor income $(1 - \tau_\ell)w_t h_{i,t} e_{i,t} \theta_{i,t}^F$, while informal workers earn $w_t h_{i,t} e_{i,t} \theta_{i,t}^I$, where w_t is the baseline wage, τ_ℓ is a fixed labor income tax rate. Let $y_t(s, e, \theta)$

be this non-financial income,

$$y_t(s, e, \theta) = \begin{cases} (1 - \tau_\ell)w_t h_t e \theta^F, & \text{if formal;} \\ w_t h_t e \theta^I, & \text{if informal;} \\ 0, & \text{if unemployed.} \end{cases}$$

Let $b_{i,t} \in \{0, 1\}$ be the *Bolsa Família* (BF) status, so that the transfer size is $T_{i,t} = T_t b_{i,t}$. At the end of each period, non-recipients in informality or unemployment enter with probability λ if eligible by an income criterium $y_{i,t} + r_t^a a_{i,t} < \bar{y}$. At the same time, beneficiaries face a probability ρ_s of being audited, where $\rho_F > \rho_I$.

We assume that audited formal recipients always leave, while informal and unemployed households stay only if eligible. Formal income is cross-checked automatically, however, the lag between entering a formal job and losing the benefit motivates a $\rho_F < 1$. Therefore, the probability of being a recipient next period is

$$\mathbb{P}[b' = 1 | s, b, \theta, e, a] = q(s, b, \theta, e, a) = \begin{cases} \lambda \cdot \mathbb{1}\{s \neq F, y + ra < \bar{y}\}, & \text{if } b = 0 \\ (1 - \rho_s) + \rho_s \mathbb{1}\{s \neq F, y + ra < \bar{y}\}, & \text{if } b = 1 \end{cases}.$$

However, we assume that assets are hidden from the government, so the agency approves within a group of observationally identical households, considering their productivities $x = (\theta, e)$. Thus, the agency approves with a conditional probability $\bar{Q}(x, b)$, weighted by the assets distribution, given by

$$\bar{Q}(x, b) = \mathbb{E}[q(x, b, a) | x].$$

Households save into a mutual fund of liquid assets $a_{i,t}$ with ex-post real return $1 + r_t^a$. They also receive lump-sum transfers τ_t by the government. Their budget constraint at time t reads:

$$c_{i,t} + a_{i,t} = (1 + r_t^a)a_{i,t-1} + y_{i,t}(s, e, \theta) + T_t b_{i,t} + \tau_t.$$

Incomplete markets impose a borrowing constraint $a_{i,t} \geq \underline{a}$. Frictions in the labor market imply formal households delegate their labor supply decisions to labor unions, and thus take worked hours as given, so their labor disutility is internalized at the union level, not by individual households. While, informal workers choose hours to maximize their utility, thus, for each i , the Intratemporal Condition gives us

$$\psi h^{1/\varphi} = w e \theta^I u'(c) \quad \Leftrightarrow \quad h^I(c, e, \theta) = \left[\frac{w e \theta^I c^{-1/\gamma}}{\psi} \right]^\varphi.$$

In order to avoid high-productivity workers going to informality to work unrealistic hours, we impose a rigidity on informals hours, where informal hours are capped at the

formal full time h^F , so that $h_{i,t}^I = \min \{h^I(c, e, \theta), h^F\}$. This is motivated by Figure 4.

Such as Esteban-Pretel and Kitao (2021) and Finamor (2025), given a sector s , we assume that each worker faces an exogenous layoff probability δ_s , and receives an offer from s with probability π_s if employed or π_s^U if unemployed. The household accepts the offer with the higher value, smoothed by additive idiosyncratic preference shocks explained later.

The Bellman Equation for each individual i in sector $s \in \{F, I\}$ is

$$V_t^s(a, e, \beta, \theta) = \max_{c, h, a'} \left\{ \frac{c^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} - \psi \frac{h^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}} + \beta \mathbb{E}_t \left[\delta_s V_{t+1}^U(\cdot) + (1-\delta_s) \left(\begin{aligned} &\pi_F(1-\pi_I) \max \{V_{t+1}^s(\cdot, \theta), \mathbb{E}_\theta V_{t+1}^F(\cdot, \theta')\} \\ &+ (1-\pi_F)\pi_I \max \{V_{t+1}^s(\cdot, \theta), \mathbb{E}_\theta V_{t+1}^I(\cdot, \theta')\} \\ &+ \pi_F\pi_I \max \{V_{t+1}^s(\cdot, \theta), \mathbb{E}_\theta V_{t+1}^I(\cdot, \theta'), \mathbb{E}_\theta V_{t+1}^F(\cdot, \theta')\} \\ &+ (1-\pi_F)(1-\pi_I)V_{t+1}^s(\cdot, \theta) \end{aligned} \right) \right] \right\}$$

s.t. $c + a' = (1 + r_t^a)a + y_t(s, e, \theta) + T_t b + \tau_t$
 $a' \in [a, \infty),$

where formal workers take hours as given. And for each unemployed worker is

$$V_t^U(a, e, \beta) = \max_{c, a'} \left\{ \frac{c^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} + \beta \mathbb{E}_t \left[(1-\pi_F^U)(1-\pi_I^U)V_{t+1}^U(\cdot) \right. \right. \\ \left. \left. + \pi_F^U(1-\pi_I^U) \max \{V_{t+1}^U(\cdot), \mathbb{E}_\theta V_{t+1}^F(\cdot)\} \right. \right. \\ \left. \left. + (1-\pi_F^U)\pi_I^U \max \{V_{t+1}^U(\cdot), \mathbb{E}_\theta V_{t+1}^I(\cdot)\} \right. \right. \\ \left. \left. + \pi_F^U\pi_I^U \max \{V_{t+1}^U(\cdot), \mathbb{E}_\theta V_{t+1}^I(\cdot), \mathbb{E}_\theta V_{t+1}^F(\cdot)\} \right] \right\}$$

s.t. $c + a' = (1 + r_t^a)a + y_t(U) + T_t b + \tau_t$
 $a' \in [a, \infty).$

As in Caliendo, Dvorkin, and Parro (2019), before deciding, the households receive an offer $A \in \{F, I, FI\}$ and draws a vector $\{\varepsilon^j\}$ over the options $j \in \mathcal{S}$, with scale σ_S ¹, and picks the option j maximizing $V^j + \varepsilon^j$. This delivers a closed-form expected value,

$$\mathbb{P}^A(j) = \frac{\exp(V^j/\sigma_S)}{\sum_{k \in \mathcal{S}} \exp(V^k/\sigma_S)}, \quad \mathbb{E} \left[\max_{j \in \mathcal{S}} \{V^j + \varepsilon^j\} \right] = \sigma_S \ln \sum_{j \in \mathcal{S}} \exp \left(\frac{V^j}{\sigma_S V_a} \right)^2.$$

¹ σ_S governs the smoothness of the decision. As $\sigma_S \rightarrow 0$ it collapses to the deterministic max.

² σ_S is scaled by the marginal value of wealth, V_a , so that the randomness of the sectoral choice does not vary mechanically with wealth. We use a standard linear weighted approximation for the log-exp sum.

These shocks render the transition a differentiable function, which allows it to be linearized when computing the sequence-space Jacobians, as in Iskhakov et al. (2017). Therefore, the sector transition matrix $\Pi_t^s = \mathbb{P}(s'|s)$ is given by

$$\Pi_t^s = \begin{bmatrix} (1 - \delta_F)\phi_F(F) & (1 - \delta_F)\phi_F(I) & \delta_F \\ (1 - \delta_I)\phi_I(F) & (1 - \delta_I)\phi_I(I) & \delta_I \\ \phi_U(F) & \phi_U(I) & \phi_U(U) \end{bmatrix},$$

where $\phi_s(s')$ is the probability of accepting an offer s' given that the worker is in sector s ,

$$\begin{aligned} \phi_F(I) &= \pi_F \pi_I \mathbb{P}^{FI}(I) + (1 - \pi_F) \pi_I \mathbb{P}^I(I), & \phi_F(F) &= 1 - \phi_F(I), \\ \phi_I(F) &= \pi_F \pi_I \mathbb{P}^{FI}(F) + \pi_F (1 - \pi_I) \mathbb{P}^F(F), & \phi_I(I) &= 1 - \phi_I(F), \\ \phi_U(F) &= \pi_F^U \pi_I^U \mathbb{P}^{FI}(F) + \pi_F^U (1 - \pi_I^U) \mathbb{P}^F(F), \\ \phi_U(I) &= \pi_F^U \pi_I^U \mathbb{P}^{FI}(I) + (1 - \pi_F^U) \pi_I^U \mathbb{P}^I(I), & \phi_U(U) &= 1 - \phi_U(F) - \phi_U(I). \end{aligned}$$

We denominate the steady-state size of each sector $s \in \{F, I, U\}$ as α_s , such that $\alpha_F + \alpha_I + \alpha_U = 1$. Let $\Lambda_t(s, b, \theta, \beta, e, a)$ be the household distribution over a given period of time.

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0, 1]$ intermediate inputs produced by formal firms,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{j,t} = Z_t l_{j,t}$. They hire labor at real wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu W_t / Z_t$, and real wage equals $w_t = Z_t / \mu$ at every date in steady-state. There is a mass 1 of outstanding firm shares with end-of-period price p_t^e (Auclert, Rognlie, and Straub 2025).

We assume sticky prices following Rotemberg (1982), standard in Neo-Keynesian's literature, with a quadratic adjustment cost

$$\Psi_t(\pi_t) = \frac{\mu}{\mu - 1} \frac{1}{2\kappa} [\log(1 + \pi_t)]^2 Y_t.$$

Firms' dividends are taxed at rate τ_d , so $d_t = (1 - \tau_d)(Y_t - w_t L_t) - \Psi_t(\pi_t)$, where $L_t = \mathbb{E}[l_{j,t}]$ is the aggregate formal labor. Such as Hagedorn, Manovskii, and Mitman

(2019), the Phillips curve for price inflation is given by

$$\ln(1 + \pi_t) = \kappa \left[\frac{w_t}{Z_t} - \frac{1}{\mu} \right] + \frac{\ln(1 + \pi_{t+1})}{1 + r_{t+1}} \cdot \frac{Y_{t+1}}{Y_t},$$

Additionally, there is a representative informal firm operating in a perfectly competitive market and producing under constant returns to scale, so $Y_t^I = w_t L_t^I$.

Unions

Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to Rotemberg (1982)'s adjustment costs. We assume that unions allocate all formal labor hours uniformly across agents, so $h_{i,t} = h_t$ (Auclert, Rognlie, and Straub 2025).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $\bar{C}_t = \int_0^1 c_{it} di$, discounted by the average discount factor $\bar{\beta}$, considering only formal workers (Auclert, Rognlie, Souchier, et al. 2021). This results in a Phillips curve for wage inflation $\pi_t^w = (1 + \pi_t)w_t/w_{t-1}$ given by

$$\ln(1 + \pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} - \frac{(1 - \tau_\ell)w_t}{\mu_w F_t} \int_F e_{i,t} \theta_{i,t} u'(c_{i,t}) d\Lambda_t \right] + \bar{\beta} \ln(1 + \pi_{t+1}^w).$$

The Phillips Curve calculations for price and wage are found in the Appendix.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r_t^* + \phi \pi_t$, where r^* is the neutral real interest rate, i_t is the nominal interest rates, π_t is the inflation, and $\phi > 1$ (Taylor 1993). Let $1 + r_t = (1 + i_{t-1})/(1 + \pi_t)$ be the real interest rates.

In the governmental sphere, we have a government that collects taxes, makes transfers, and incurs debt; therefore, the government's budget constraint is characterized by

$$\begin{aligned} \tau_t + T_t \cdot \text{BF}_t + (1 + r_t)B_{t-1} &= B_t + \tau_\ell w_t L_t + \tau_d(Y_t - w_t L_t) \\ &= B_t + \tau_d Y_t - (\tau_d - \tau_\ell)w_t L_t. \end{aligned}$$

We consider two types of government policy, a revenue-neutral and a deficit-financed. The policymaker chooses a path of the conditional transfers and adjusts transfers or debt such that the budget is balanced at each period.

The revenue-neutral policy sets a path of T_t and adjusts transfers such that

$$\tau_t = \tau_d Y_t - (\tau_d - \tau_\ell)w_t L_t - T_t \cdot \text{BF}_t - r_t \bar{B},$$

which means that the fiscal policy is passive and the monetary policy is active. The nominal interest rate adjust at least one-for-one to changes in inflation.

However, governments normally finance themselves by increasing deficit and issuing new debt. The deficit-financed policy sets a path of T_t and τ_t , adjusting the debt to pay off. We assume that the government repays its debt in the long run by reducing transfer (when $\tau_t > 0$) or increasing taxes (when $\tau_t < 0$), with elasticity ϕ_B ,

$$\tau_t = \bar{\tau} - \phi_B \cdot (B_{t-1} - \bar{B}).$$

Or, in another words, using Auclert, Rognlie, and Straub (2025) notation, the debt persistency is defined by

$$dB_t = (1 + r_t)dB_{t-1} + d\tau_t = (1 + r - \phi_B)dB_{t-1} = \rho_B dB_{t-1},$$

where ρ_B is the debt persistency, which we calibrate so that debt has a half-life of 5 years, so $\rho_B = 0.5^{1/20} \approx 0.966$ and $\phi_B = 0.047$.

This situation leads to larger multipliers and great inequality (Auclert, Rognlie, and Straub 2023). In the presence of nominal rigidity and non-Ricardian consumer behavior, Angeletos, Lian, and Wolf (2024) show that fiscal deficits of this sort self finance, so $\lim_{t \rightarrow \infty} B_t = \bar{B}$.

Finally, there is a mutual fund collecting household savings $a_{i,t}$ and investing on government bonds and the stock market. No-arbitrage condition garanties that

$$(1 + r_{t+1})p_t^e = p_{t+1}^e + (1 - \tau_d)d_{t+1} \quad \Leftrightarrow \quad p_t^e = \sum_{s \geq 1} \frac{(1 - \tau_d)d_{t+s}}{\prod_{k=1}^s (1 + r_{t+k})},$$

which implies

$$1 + r_t^a = \frac{(1 + r_t)B_{t-1} + d_t + p_t^e}{B_{t-1} + p_{t-1}^e} = 1 + r_t, \quad \forall t \geq 1.$$

At $t = 0$ the shock is unanticipated, so households hold a portfolio bought at steady-state prices. The initial return r_0^a is such that the liquidation value of the mutual fund equals the beginning-of-period value of its stock and bond holdings,

$$(1 + r_0^a)A = p_0^e + d_0 + (1 + r_0)B,$$

so $1 + r_0^a$ is a weighted average of the initial stock return $(p_0^e + d_0)/p_0^e$ and the bond return $1 + r_0$, with weights given by the steady-state portfolio shares p^e/A and B/A . The shock thus revalues household wealth on impact through both assets, while every Euler equation prices the ex-ante return $r_{t+1}^a = r_{t+1}$.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). First, the asset market reaches equilibrium when household savings equal stocks and government bonds

$$A_t = \int a_{i,t} d\Lambda_t(s, b, \theta, \beta, e, a) = B_t + p_t^e;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$L_t = \int \mathbb{1}\{s_{i,t} = F\} \cdot \theta_{i,t}^F \cdot e_{i,t} \cdot h_{i,t} d\Lambda_t(.);$$

$$L_t^I = \int \mathbb{1}\{s_{i,t} = I\} \cdot \theta_{i,t}^I \cdot e_{i,t} \cdot h_{i,t} d\Lambda_t(.);$$

and the goods market clear when $Y + Y^I = C + \Psi_t$. We set the goods market to clear by Walras' Law. Lastly, the BF benefit is given to all eligible households,

$$\text{BF}_t = \int b_{i,t} d\Lambda_t(.).$$

Equilibrium. Given initial conditions $(B_{-1}, p_{-1}^e, \Lambda_0)$ and exogenous paths $\{T_t, r_t^*\}_{t \geq 0}$, an equilibrium is a sequence of prices $\{w_t, \pi_t, \pi_t^w, i_t, r_t, r_t^a, p_t^e\}$, macroeconomic aggregates $\{Y_t, Y_t^I, L_t, L_t^I, h_t, d_t, \tau_t, B_t, C_t, A_t, \text{BF}_t\}$, transition matrices $\{\Pi_t\}$, household policies $\{c_t, a_t, h_t^I\}$ over the state $(s, b, \theta, \beta, e, a)$, and distributions $\{\Lambda_t\}$ such that, for all $t \geq 0$:

1. The policies solve the Bellman equations of each sector;
2. Firms price according to the Price Phillips curve and the union sets wages according to the Wage Phillips curve, with $h_{i,t} = h_t$ for every formal worker;
3. The government satisfies its budget constraint and fiscal rule, and monetary policy follows the Taylor rule, with $1 + r_t = (1 + i_{t-1})/(1 + \pi_t)$; The mutual fund prices stocks by no-arbitrage and pays r_t^a ;
4. $A_t = B_t + p_t^e$, L_t and L_t^I equal the effective labor supplied by formal and informal workers, $\text{BF}_t = \int b d\Lambda_t$, and $Y_t + Y_t^I = C_t + \Psi_t$ holds by Walras' law.

A steady state is an equilibrium in which prices and aggregates are constant, Λ is invariant, and Π is a fixed point. Numerically, the transition solves for $\{B_t, L_t, h_t, \pi_t, w_t, \tau_t, p_t^e\}$ such that the fiscal rule, the asset and labor markets, both Phillips curves, the government budget and the no-arbitrage condition hold.

2. Calibration

Data

To calibrate this model, we use Brazilian data and standard parameters in HANK Literature. Most of the data refers to their average values in 2025.

The labor market moments were obtained in *Pesquisa Nacional por Amostra de Domicílio Contínua* (PNAD-C), an annual survey reporting formality status, wages, weekly worked hours, etc. This panel is interesting to match the sector transitions and the worked hours and wages, for both formal and informal households. However, since it is a survey, it might have under-reporting of informal households.

Some general macroeconomic measures were also used, such as the Government Debt (in the Brazilian National Treasury), the interest rates, and the inflation, available in the *Índice Brasileiro de Geografia e Estatística* (IBGE). The Gini coefficient of wealth was obtained from the 2025 Global Wealth Report, from UBS, and the Wealth Inequality Statistics were obtained from the 2024 Wealth Inequality Database³. *Bolsa Família* data is available in *Cadastro Único* (CadÚnico), in VIS Data⁴, a platform with several statistics about the benefit.

Additionally, Annual PNAD-C has information about whether a individual is beneficiary of the benefit. We identify each household and consider every individual living in a beneficiary household as receiving the benefit. Among those, we calculate the mean reported benefit T and the mean share of beneficiaries.

Table 1: Bolsa Família 2025: Statistics

	PNAD	VisData	Ratio
Households (M)	13.64	18.71	1.37
Individuals (M)	46.60	48.92	1.05
People/HH	3.42	2.61	0.77
Value (R\$/month)	669.55	710.65	1.06

The rule to receive BF is based on a R\$ 218 threshold for the family income per capita, therefore there are some formal individuals receiving the benefit. Also, CadÚnico normally delay to remove the benefit for non-eligible individuals, which can explain why some formal individuals receive BF.

In order to construct quarter-to-quarter transition probabilities across the three sectors $s \in \{F, I, U\}$, we reconstruct the PNAD-C rotating panel using Osório (2022), which estimates the dwelling and individual identifiers in the survey. The estimation of the pandemic years is not reliable.

³Available in [WID World](#).

⁴Available in [VIS DATA 3](#) and [Cadastro Único](#).

Table 2: Bolsa Família 2025: Coverage

	%
BF / Labor Force	16.1
BF in Formal	7.2
BF in Informal	23.9
BF in Unemployed	33.6
BF in Non-Participating	21.8
Spending (Labor Force) / Wage Bill	2.1
$T/\mathbb{E}(y^F)$	10.0
$\bar{y}_{law}/\mathbb{E}(y^F)$	17.6

The formality status is defined as (i) formal workers have signed work card, are militaries, or employers, and (ii) informal workers do not have a signed work card, are self-employed, or unpaid family. We only consider household in the labor force, dropping those who are non-participating, retired, or students, to keep the empirical states aligned with the model's.

Sectoral Transitions

The transition matrix across $s \in \{F, I, U\}$ is estimated on the PNAD-C rotating panel, which follows each household for five consecutive quarters. Two issues arise before the matrix can be taken to the model.

First, quarter-to-quarter transitions are contaminated by classification error: a worker whose contract status is misreported in one interview appears as two spurious moves. The error is detectable because a Markov chain must satisfy $P(h) = P(1)^h$. Table xxx compares the matrix estimated directly at horizons of one to four quarters with the iterated one-quarter matrix. At one year the quarterly matrix predicts $\mathbb{P}(F \rightarrow I) = 0.251$, while the same panel measures 0.116; the same gap appears for $\mathbb{P}(I \rightarrow F)$, 0.334 against 0.155. Iterating $P(1)$ compounds the measurement error, so the one-quarter matrix overstates mobility by a factor of two.

We therefore estimate the transition matrix over one year, from the first to the fifth visit, and convert it to the quarterly frequency by its principal fourth root,

$$\Pi_s = P(4)^{1/4},$$

computed by eigendecomposition. The root is a valid transition matrix here: all entries are non-negative and the rows sum to one, so no regularization of the generator is needed. The resulting quarterly flows are $\mathbb{P}(F \rightarrow I) = 9.3\%$ and $\mathbb{P}(F \rightarrow U) = 1.9\%$, against 0.084 and 0.012 in the raw quarterly panel. The cost of this choice is that round trips within the year are not observed, so Π_s is a lower bound on true mobility; we prefer it to a matrix whose implied one-year flows are twice what the data show.

Second, panel attrition is selective: households that stay in the sample for five visits are more likely to be formally employed. The stationary distribution of the raw matrix is (0.549, 0.415, 0.036) against the cross-sectional shares (49.8%, 44.5%, 5.6%). We correct it by scaling the columns of P by a vector c and renormalizing the rows,

$$\tilde{P}_{ss'} = \frac{P_{ss'} c_{s'}}{\sum_k P_{sk} c_k},$$

where c solves the fixed point that makes the stationary distribution of $\tilde{P}$ equal to the annual stocks α , normalized at $c_F = 1$. The tilt reweights the destinations, not the origins, so it preserves the ranking of the flows out of each state; it moves $\mathbb{P}(F \rightarrow I)$ by less than one percentage point.

Job Tenure

The tenure question (V4040, with V40401 to V40403) gives the time the worker has been in the current job, which disciplines how often jobs end. Restricting to private employees with a formal contract (VD4009 = 01), since public employment has a stability the model does not represent, mean tenure is 86.826 months and a share 0.207 of workers have been less than one year in the job. Under a constant exit rate η and stationary hiring, tenure in the stock of employed workers is geometric, so

$$\Pr(\text{tenure} < 1 \text{ year}) = 1 - (1 - \eta)^4 \implies \eta = 1 - (1 - \Pr(\text{tenure} < 1 \text{ year}))^{1/4} = 0.056.$$

The share with less than three months of tenure, 0.070, implies the same rate, which supports the constant-hazard approximation. In the model the job exit rate is $\eta = \delta_F + (1 - \delta_F) \phi$, where ϕ is the probability that an employed formal worker receives and accepts an outside offer. Since δ_F and the sectoral flows are matched by construction, the residual identifies the formal-to-formal job change, which the sectoral flows cannot see, and it is what we use to calibrate the taste dispersion σ_S . Note that this discipline is only available once the quarterly flows come from the annual matrix: with the raw quarterly flows the sectoral exit rate alone, 0.097, already exceeds the job exit rate 0.056, which is impossible since every sectoral move is also a job change.

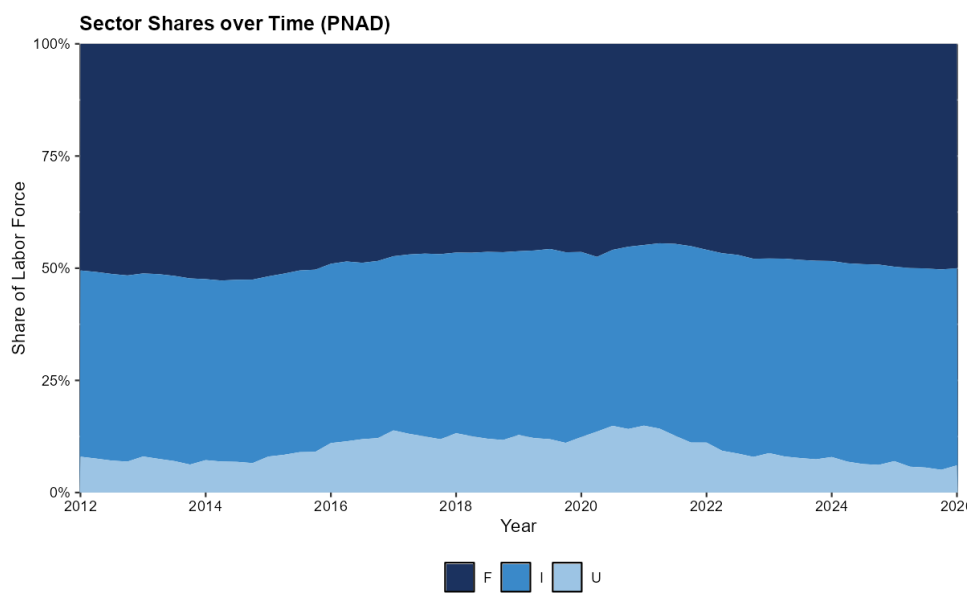


Figure 1: Share of Formal, Informal, and Unemployed Households

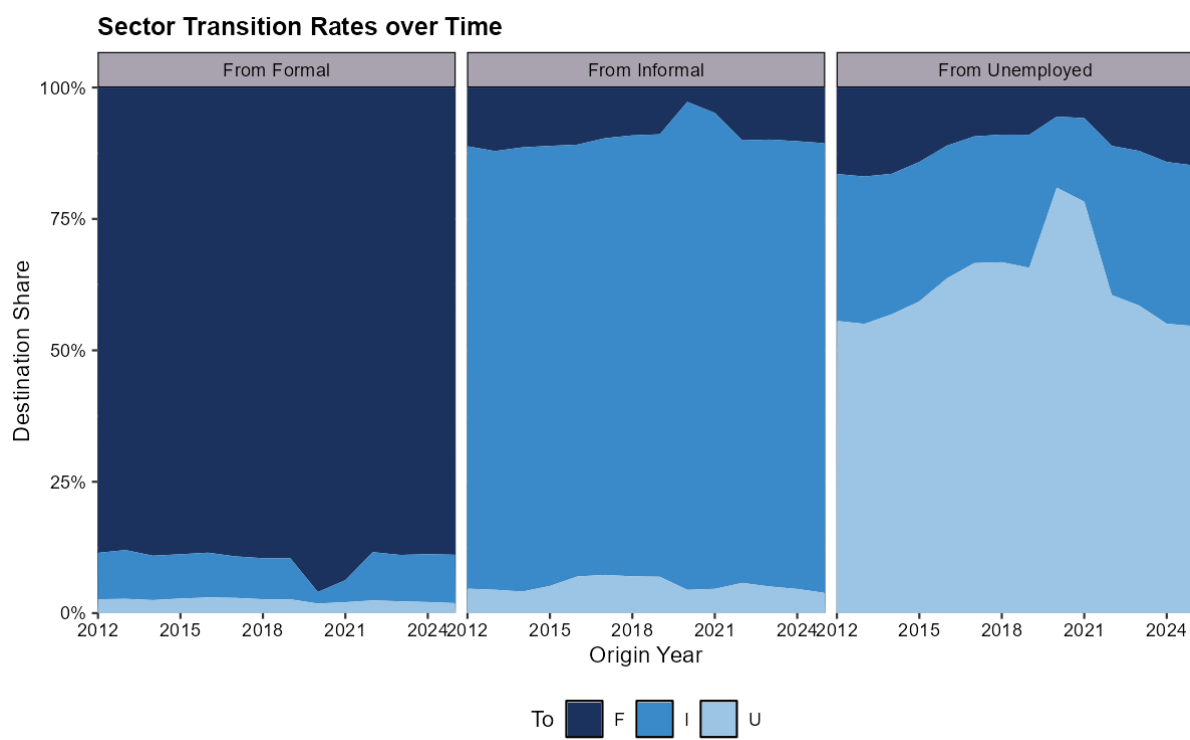


Figure 2: Sector Transitions over time

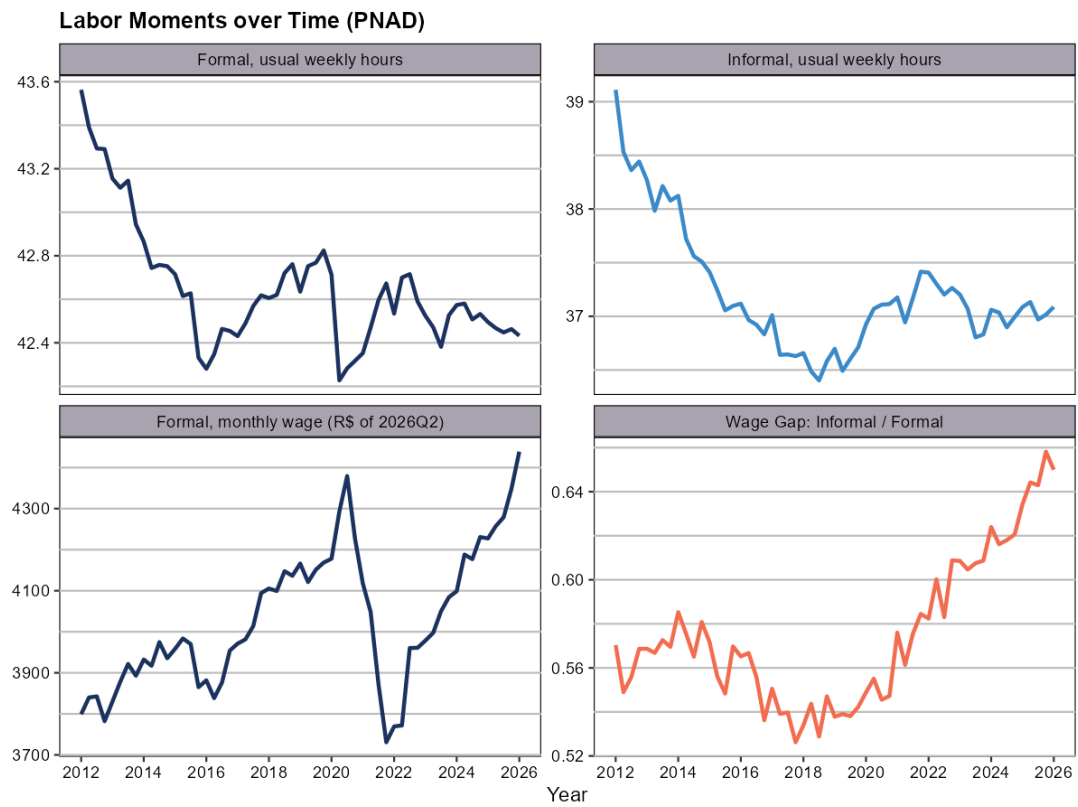


Figure 3: Wage and Worked Hours for each Sector

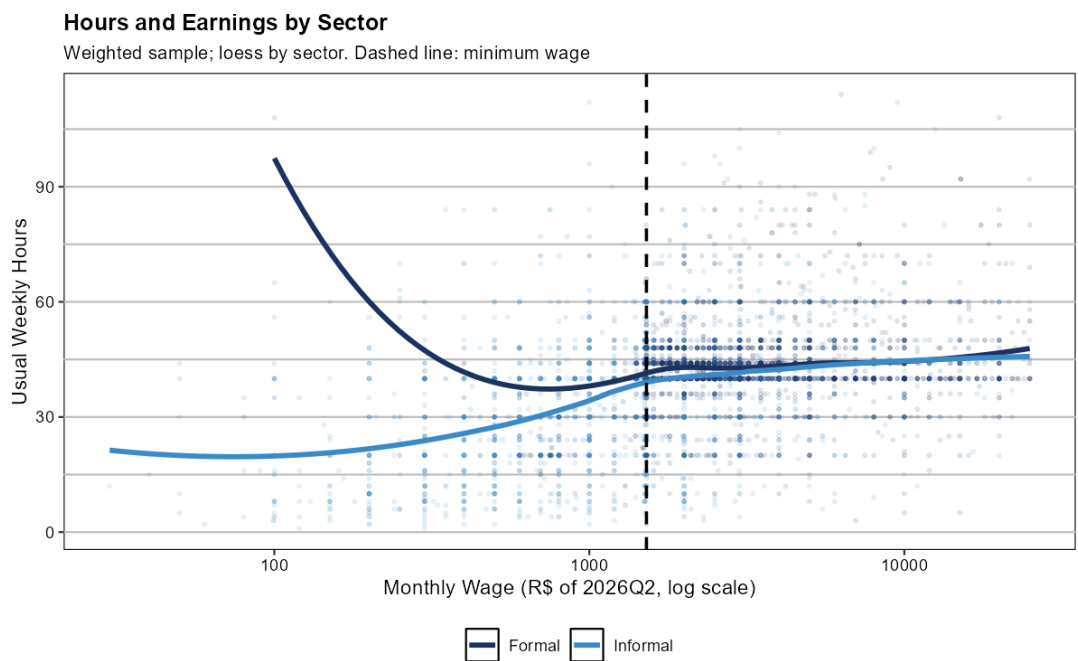


Figure 4: Average Worked Hours by Wage for each Sector

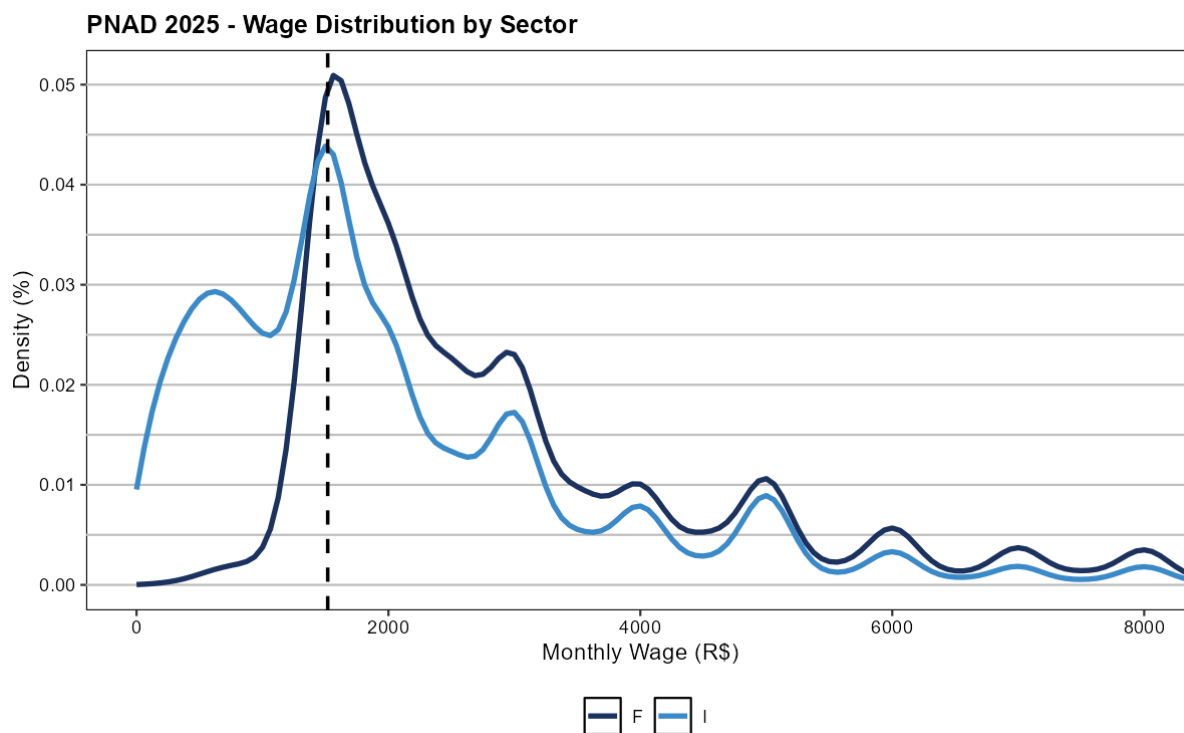


Figure 5: Wage Distribution for each Sector

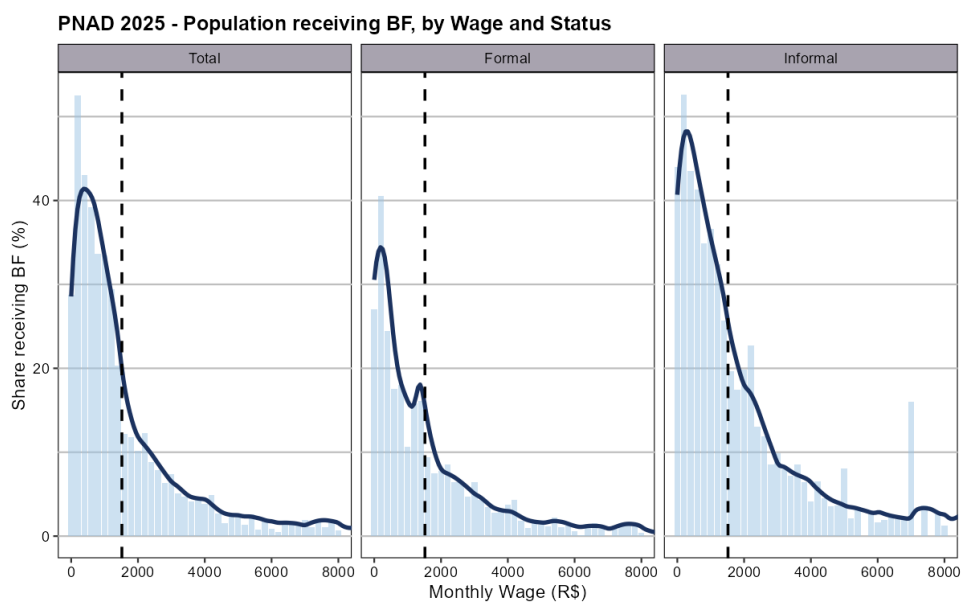


Figure 6: Share of Households Receiving Benefit by Wage Level

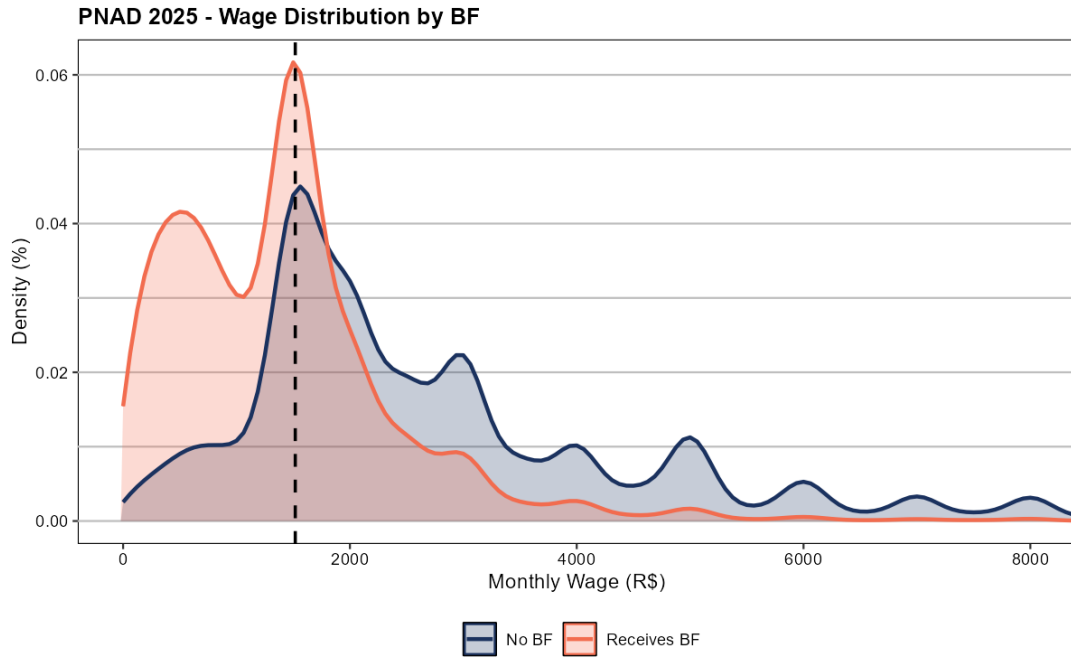


Figure 7: Wage Distribution by Benefit Status

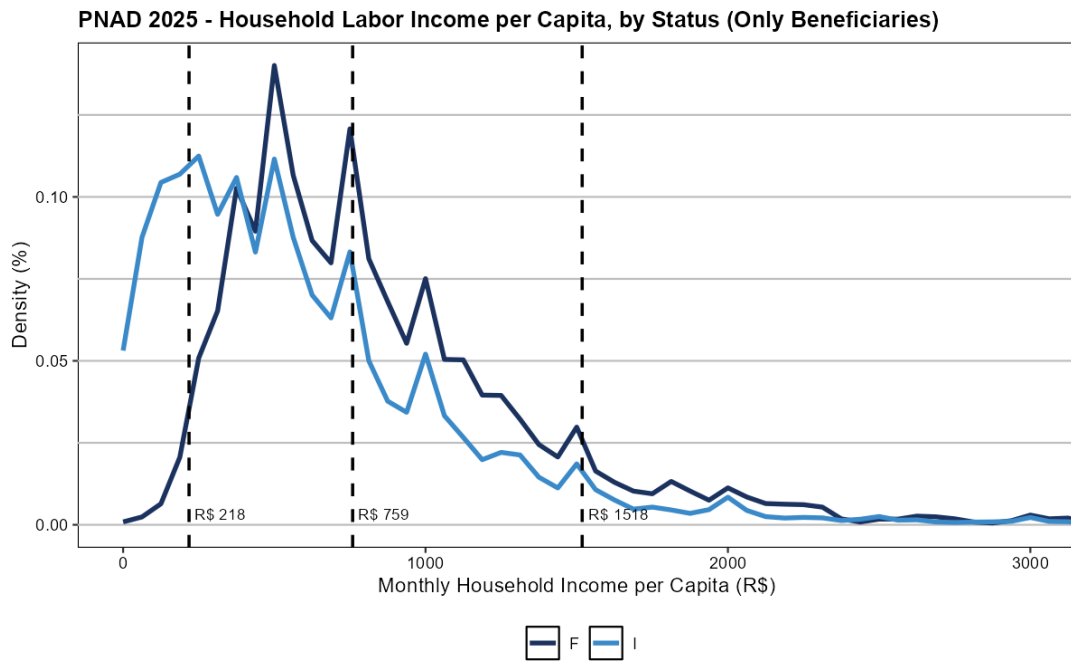


Figure 8: Familiar Wage Distribution by Sector, for Beneficiaries

Methods

The main computational methods are the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, et al. 2021), and the Endogenous Grid Method (EGM) (Carroll 2006). The

EGM was combined with the discrete choice among sectors using Iskhakov et al. (2017), with taste shocks.

The central idea of SSJ is to summarize the entire model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous ones. The Implicit Function Theorem allows us to differentiate this system and obtain General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G} d\mathbf{Z} = -\mathbf{F}_{\mathbf{X}}^{-1} \mathbf{F}_{\mathbf{Z}} d\mathbf{Z}.$$

For each shock in *Bolsa Família*, we can compute the Jacobian $dC = \mathbf{G}_{C,T} dT$ of the change in the conditional transfer on household consumption. The Jacobian is computed by perturbing the exogenous variable T and solving the model for the new equilibrium, then calculating the change in consumption C .

Further, we can compute the consumption response to a monetary shock $dC = \mathbf{G}_{C,i} di$ in the presence of the transfer, and compare it to the counterfactual. We measure the stabilization effect of the transfer as the difference between the response in the economy with the program and in the $T = 0$ counterfactual,

$$\text{Stab}(T) = dC \Big|_{T>0} - dC \Big|_{T=0},$$

read either as an impulse response or as the induced change in the volatility of consumption $\Delta \text{Var}(C_t)$, a program that smooths fluctuations delivers $\Delta \text{Var}(C_t) < 0$. This effect can be decomposed in a insurance and a composition channel,

$$d\mathbf{X} = d\mathbf{X}^{\text{ins}} + (d\mathbf{X} - d\mathbf{X}^{\text{ins}}) = \mathbf{G}^{\text{ins}} d\mathbf{Z} + (\mathbf{G} - \mathbf{G}^{\text{ins}}) d\mathbf{Z}.$$

The sectoral block of the transition matrix Π_t is endogenous and depends on the continuation values, $\Pi_t = \Pi(\mathbf{V}_t, \Lambda_t)$, where $\mathbf{V}_t = (V_t^F, V_t^I, V_t^U)$. The household's distribution evolves by combining the asset policy with the Markov transition over the states, $\Lambda_{t+1} = \mathcal{T}(\Pi_t, a_t(s, \theta, \beta, e))$. To solve the steady-state distribution, we iterate until Π_t converges to its fixed point, analogue to Aguirregabiria and Mira (2007).

The insurance channel is captured by the frozen- Π . Holding sectoral sorting fixed, the transfer relaxes the budget of unemployed and low-income informal households, and, since these are high-MPC, it dampens the aggregate response (Auclert, Rognlie, and Straub 2024).

The composition channel is captured by the moving- Π Jacobian, because the CCT raises the value of informality relative to formal work, so workers are tilted toward the informal sector, which is riskier. The insurance term is stabilizing, while the composition is potentially destabilizing. The sign and magnitude of the net effect are the quantitative answer to our question.

Letting the transition respond, $d\Pi_t = \Pi_{\mathbf{V}} d\mathbf{V}_t$, yields the general equilibrium Jacobian $\mathbf{G}$. This dynamic is obtained as the dynamic analogue of the steady-state: given a shock path, we iterate so that households' sectoral choices are consistent with the values those very choices generate along the transition (Auclert, Bardóczy, et al. 2021).

Calibration

Table 3: External Calibration

Description		Value	Source
<i>Households and Labor Market</i>			
γ	EIS	0.5	Standard in Literature
φ	Informal Frisch Elasticity	0.5	Standard in Literature
ψ	Disutility of Labor	0.927	Normalization: $h_F = 1.0$
β_H	Patient's Discount Factor	0.979	$A = B + p^e$
δ_F	Job-Loss Probability to Formal	0.019	PNAD-C Panel (Osório 2022)
δ_I	Job-Loss Probability to Informal	0.038	PNAD-C Panel (Osório 2022)
<i>Firms</i>			
w	Average Wage	1.0	Normalization, Reference
μ, μ_w	Price Markup	1.11	Basu and Fernald (1997)
κ	Price-NKPC Slope	0.10	Kaplan, Moll, and Violante (2018)
κ_w	Wage-NKPC slope	0.10	Avg. Wage Rigidity of 1 Year
<i>Government and Monetary Policy</i>			
ϕ	Taylor Rule Coefficient	1.5	Standard (Taylor 1993)
r^*	Neutral Real Interest Rate	1.2%	5% Annual Average
τ_ℓ	Labor Income Tax Rate	13.0%	Tax Payroll for Average Worker
τ_d	Profit Tax Rate	34.0%	IRPJ 25%, and CSLL 9%
B/Y	Government Debt-to-GDP Ratio	78.6%	Brazilian National Treasure
ϕ_B	Sensitivity to Pay the Debt	0.047	B half-life of 5 years
$BF \cdot T/W$	Bolsa Família Spending (%Wage Bill)	2.1%	CadÚnico, PNAD-C
ρ_I	Informal Fiscalization Probability	0.12	CadÚnico, 2y Review

Each model period stands for one quarter and the computational time horizon is 100 quarters (or 25 years). We internally calibrate the steady state by setting β_H to balance the asset market; $\pi_F, \pi_I, \pi_{UF}, \pi_{UI}$ to match the sectoral transition; and $\lambda, \bar{y}, \rho_F$ to match the BF coverage by sector in steady state (Figure 2). Also, we internally calibrate parameters using the Simulated Method of Moments (SMM) by a Global Search on a coarse grid, followed by a local polish using the Nelder-Mead algorithm. The external calibration is shown in table 3 and the endogenous calibration is in 4.

The idiosyncratic productivity follows a log-normal $AR(1)$ process, $\ln e' = \rho_e \ln e + \varepsilon^e$, with $\varepsilon^e \sim N(0, \sigma_e^2)$. This is approximated by a finite Markov chain with $N_e = 15$ grid points (Rouwenhorst 1995). We internally calibrate ρ_e and σ_e to match the autocorrelation of wages in PNAD-C $\rho_4^y := \text{Corr}(\ln y_t, \ln y_{t-4})$ and the observed wage distribution.

Table 4: Endogenous Calibration

Description		Value	Target	Model	Data	Source
<i>Simulated Methods of Moments (SMM)</i>						
β_L	Impatient's Discount Factor	0.819	Top 10%	54.5%	71.9%	WID World
ω_I	Share of Impatient Agents	50.0%	MPC	0.162	0.2	Standard
ρ_e	Persistence of Productivity	0.966	ρ_4^y	0.813	0.846	PNAD-C Panel
σ_e	Productivity Std. Dev.	0.700	q_{50}^F	-0.118	-0.526	PNAD-C
μ_I	Informal Average Productivity	-0.500	q_{50}^I	-0.826	-0.972	PNAD-C
σ_F	Formal Std. Dev. Produc.	0.300	q_{75}^F	0.256	0.062	PNAD-C
σ_I	Informal Std. Dev. Produc.	0.400	q_{75}^I	-0.064	-0.344	PNAD-C
<i>Internal Consistency</i>						
π_F	Formal Offer Probability	28.1%	$I \rightarrow F$	10.6%	10.6%	PNAD-C Panel
π_F^U	Formal Offer Prob. (Unemp.)	25.2%	$U \rightarrow F$	14.9%	14.9%	PNAD-C Panel
π_I	Informal Offer Probability	56.8%	$F \rightarrow I$	9.3%	9.3%	PNAD-C Panel
π_I^U	Informal Offer Prob. (Unemp.)	51.3%	$U \rightarrow I$	31.2%	31.2%	PNAD-C Panel
σ_S	Smoothness of Tastes	0.500	xxx	xxx	xxx	PNAD-C
λ	BF Entry Probability	0.21	BF_U	33.6%	33.6%	PNAD-C
$\bar{y}$	BF Threshold for Informals	0.27	BF_I	23.9%	23.9%	PNAD-C
ρ_F	Formal Fiscalization Prob.	0.27	BF_F	7.2%	7.2%	PNAD-C

We also assume a log-spaced grid for assets $[0, 100]$ with $N_a = 100$ points, sufficient to cover the bottom and the top of distribution, and a log-normal sector productivity grid $\ln \theta^s \sim N(\mu_s, \sigma_s^2)$, that is discretized by Equiprobable Bins with $N_\theta = 5$ nodes.

Each household has a $q = 0.0$ probability of drawing a new discount factor (β_L, β_H) , corresponding to an average type duration of 25 years, used to capture a generation (Auclert, Rognlie, and Straub 2025). This aims to capture similar results in relation to a life-cycle overlapping generations with permanent ability shock model.

In the household preferences, elasticity of intertemporal substitution is standard in literature $\gamma = 0.5$ ⁵. The disutility of labor $\psi = 0.927$ is calibrated to normalize formal hours $h_F = 1.0$ in steady state. Finally, the preference shock smoothing parameter is set to $\sigma_S = 0.500$.

The patient's discount factor $\beta_H = 0.979$, balances the asset market in steady state. We estimate the impatient's discount factor, $\beta_L = 0.819$, and the share of impatient households, $\omega_I = 50.0\%$, to match the Gini Coefficient on Wealth (0.82, from UBS Global Report) and the income-weighted Marginal Propensity of Consumption (MPC)⁶ (0.2), letting the Top 10% Share of Wealth (71.9%, from World Inequality Database) and the share of Hand-to-Mouth households (35.0%) untargeted.

The layoff probabilities are equal to the transition probabilities in PNAD-C, $\delta_F = 0.019$ and $\delta_I = 0.038$. Further, the job offer probabilities are calibrated to match sector sizes $s \in \{F, I, U\}$ in equilibrium, and their transition, as shown in Table 5.

We internally calibrate moments of θ to match three quantiles $\{0.25, 0.50, 0.75\}$ of the

⁵EIS $\gamma = 0.5$ is consistent with micro estimates surveyed by Valickova, Havranek, and Horvath (2015).

⁶Income-Weighted MPC is consistent with the estimates of Costa-Junior, Gadelha, and Garcia-Cintado (2025) and Galvêas (2026).

Table 5: Quarterly Labor-Market Transitions (%)

Origin	Model			PNAD-C		
	F	I	U	F	I	U
Formal	88.81	9.32	1.87	88.81	9.32	1.87
Informal	10.64	85.61	3.75	10.64	85.61	3.75
Unemployed	14.93	31.25	53.82	14.93	31.25	53.82
Stationary Share	49.84	44.52	5.63	49.84	44.52	5.63

observed wage distribution for formal and informal workers, with respect to $\mathbb{E}(y^F)$, which yields $\mu_I = -0.500$, $\sigma_F = 0.300$, and $\sigma_I = 0.400$, after normalizing $\mathbb{E}[\theta_{i,t}^F] = 1$.

The average wage is normalized $w = 1.0$, so that we can reference the model as a percentage of the Wage Bill. That was preferred rather than normalizing the GDP, because there is no capital in this model, therefore the wage is a better measure of the economy's size. We defined $\mu = 1.11$, which is standard in the HANK literature (Kaplan, Moll, and Violante 2018; Auclert, Rognlie, and Straub 2025)⁷, and $\kappa = \kappa_w = 0.10$, similar to Kaplan, Moll, and Violante (2018) and consistent with an average duration of 4 quarters for readjustment⁸.

Taylor Rule is pretty standard, with $\phi = 1.5$ and $r^* = 1.2\%$, consistent with a 5% annual real interest rate for Brazilian economy (Santos and Muinhos 2024). Also, we set $\phi_B = 0.047$ so that the half-life of the debt is 5 years, slower than Auclert, Rognlie, and Straub (2025) and consistent with Campos and Cysne (2019).

Wage tax is based on the average formal worker contribution, $\tau_\ell = 13.0\%$, and profit's tax is based on the statutory rates for Brazil, IRPJ and CSLL, which are 25% and 9% respectively. The Government debt is set to 78.6% of yearly GDP, consistent with Brazilian National Treasury data. The steady state stock market size is $p^e = d/r = 3.805$, so $A/Y = 1.684$, compared to 1.167 from BCB Credit Statistics for liquid wealth.

We calibrated the *Bolsa Família* transfer size $T = 0.126$ to match the *Bolsa Família* expenditure as a percentage of the Wage Bill, $BF \cdot T/W$ from Brazilian economy. In order to keep the internal consistency of the model, we consider T as the average transfer received only by individuals in labor force, disconsidering children and elderly.

On the other hand, the entry probability $\lambda = 0.21$, the threshold $\bar{y}/y^F = 0.27$, and the audit probability for formals $\rho_F = 0.27$ are internally calibrated to match the coverage by sector, $BF_U = 33.6\%$, $BF_I = 23.9\%$, and $BF_F = 7.2\%$, where $BF_s := \mathbb{P}[b = 1|s]$. The audit probability for informals $\rho_I = 0.12$ is set by a 2 years time for review, standard in CadÚnico. $\bar{y}$ is consistent with the 35th percentile of formal income (Ferriere and Navarro

⁷Markup $\mu = 1.11$ is also consistent with the estimates of Basu and Fernald (1997) for US. De Loecker, Eeckhout, and Mongey (2021) show that the Brazilian markup is similar to the US.

⁸We calculate κ following the standard Calvo formula $\kappa = \frac{1}{1+\Gamma} \frac{(1+\theta)(1-\beta\theta)}{\theta}$, with the real rigidity coefficient $\Gamma = 5$ and the quarterly wage adjustment probability $1 - \theta = 25.4\%$ from Caccioli (2026).

2025), $y_{35}/y^F = 0.36$, and ρ_F is consistent with an average time of 11 months to lose the benefit.

3. Results

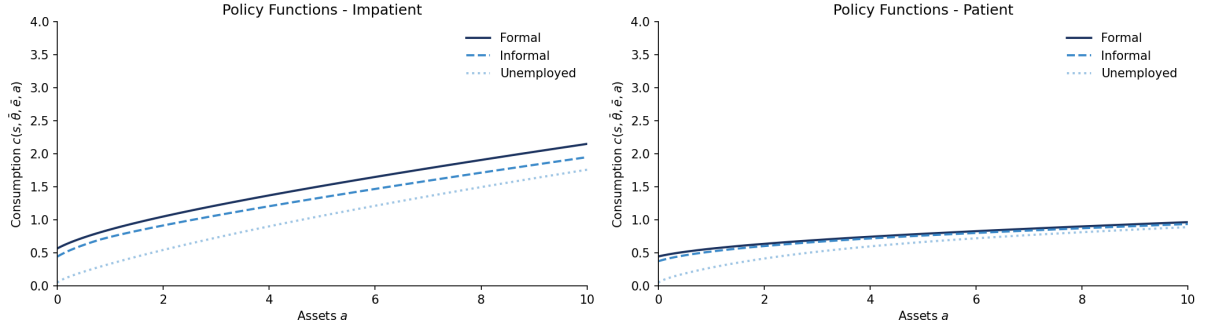


Figure 9: Steady-State Consumption Policy Function

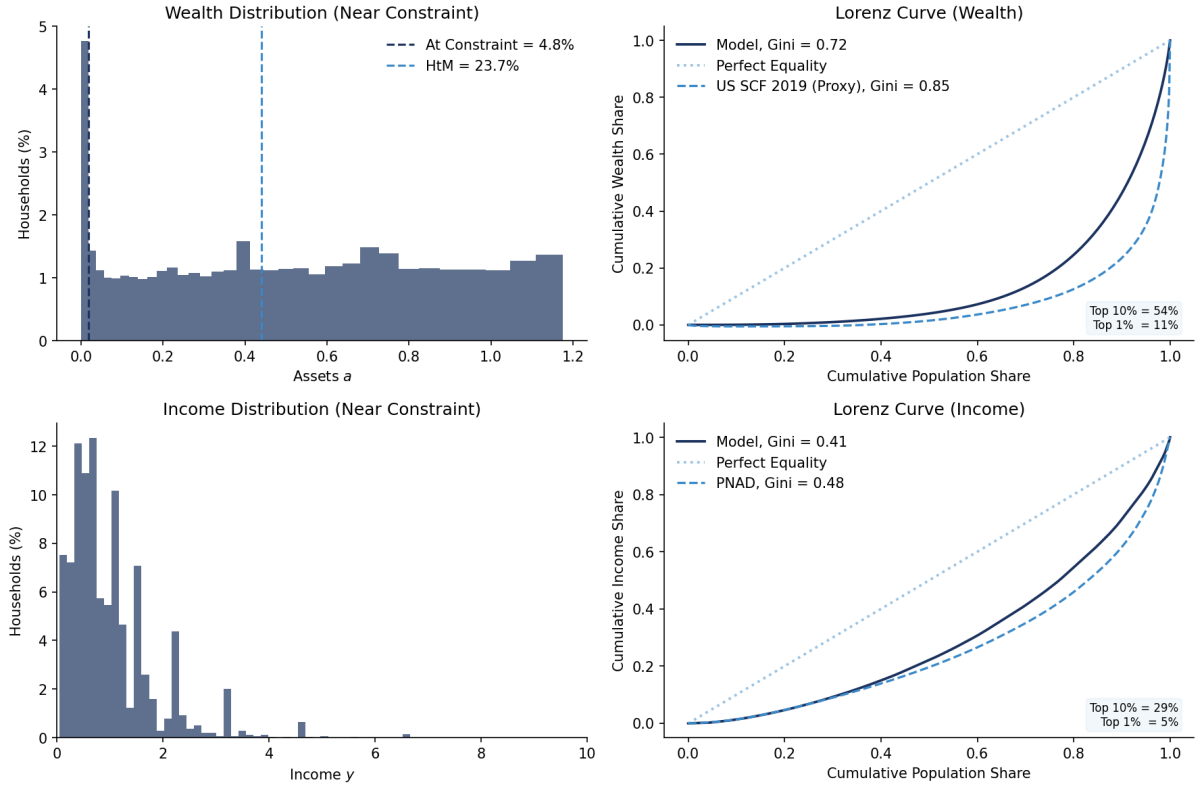


Figure 10: Wealth Distribution

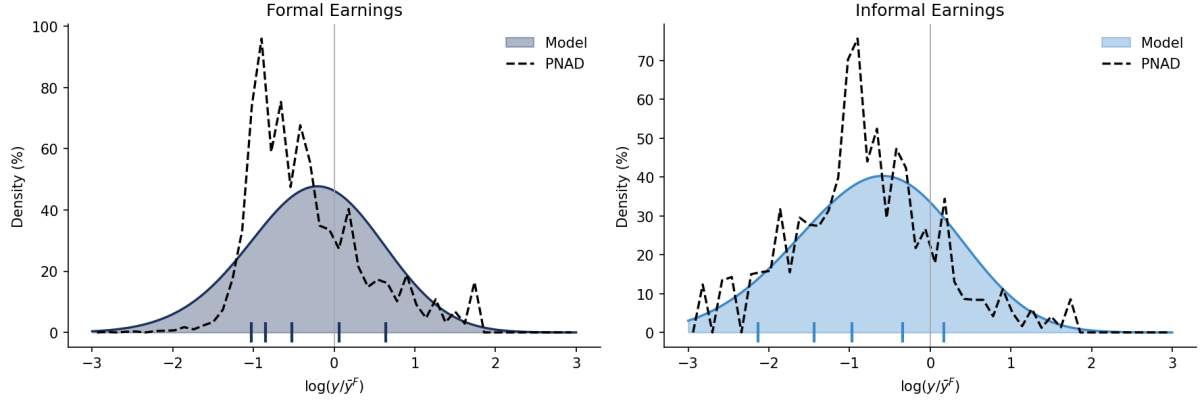


Figure 11: Income Distribution

Table 6: Steady State: Bolsa Família vs. No-Transfer Counterfactual

	With BF	No BF	Δ
Formal Share α_F	49.8%	68.1%	-18.3
Informal Share α_I	44.5%	26.6%	+17.9
Unemployment α_U	5.6%	5.2%	+0.4
Hand-to-Mouth	23.7%	31.3%	-7.6
Wealth Gini	0.719	0.739	-0.020
Wealth Share, Bottom 50%	4.0%	2.7%	+1.3
Wealth Share, Top 10%	53.7%	55.1%	-1.4
Consumption Gini	0.342	0.342	+0.000
Consumption C	1.092	1.139	-0.047
BF Spending	0.021	0.000	+0.021
Lump-Sum τ	0.046	0.112	-0.066
Welfare $\mathbb{E}[V]$, $\Delta = \text{CEV}$	-34.463	-32.476	-5.76%

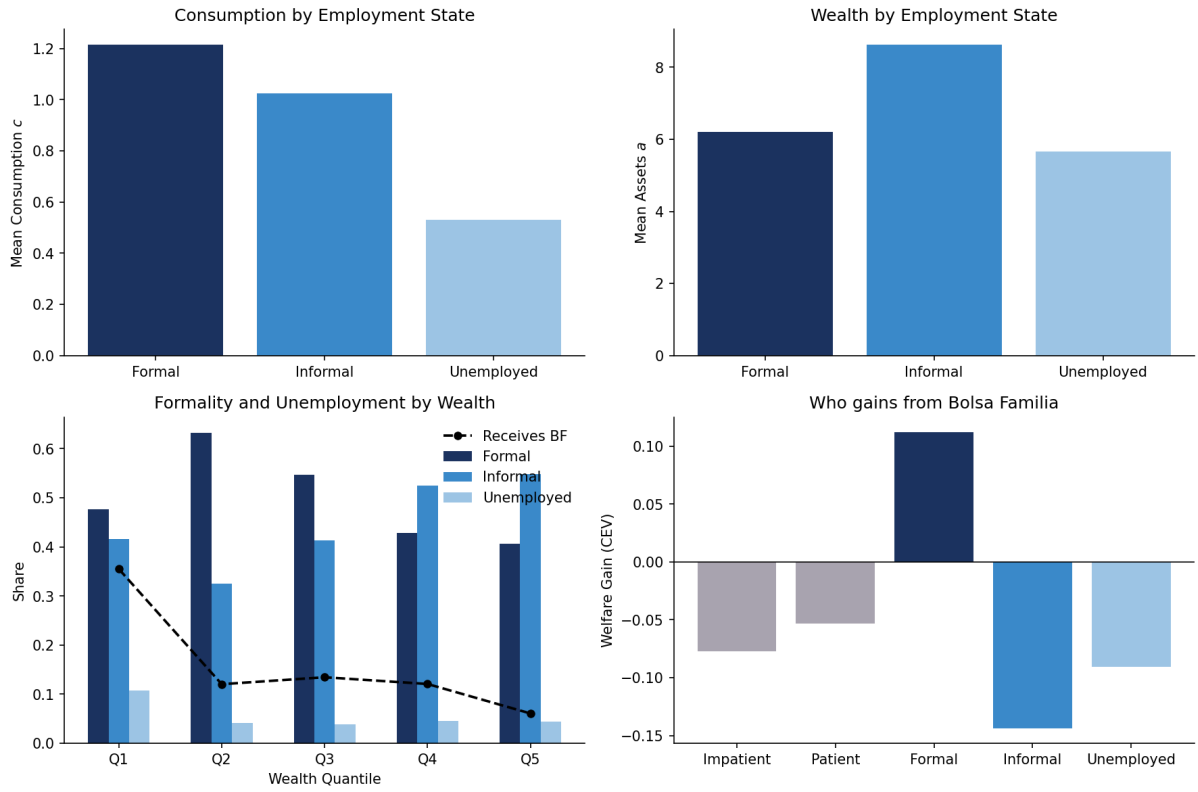


Figure 12: Steady-State Descriptive Statistics



Figure 13: Steady-State Sensitivity Analysis

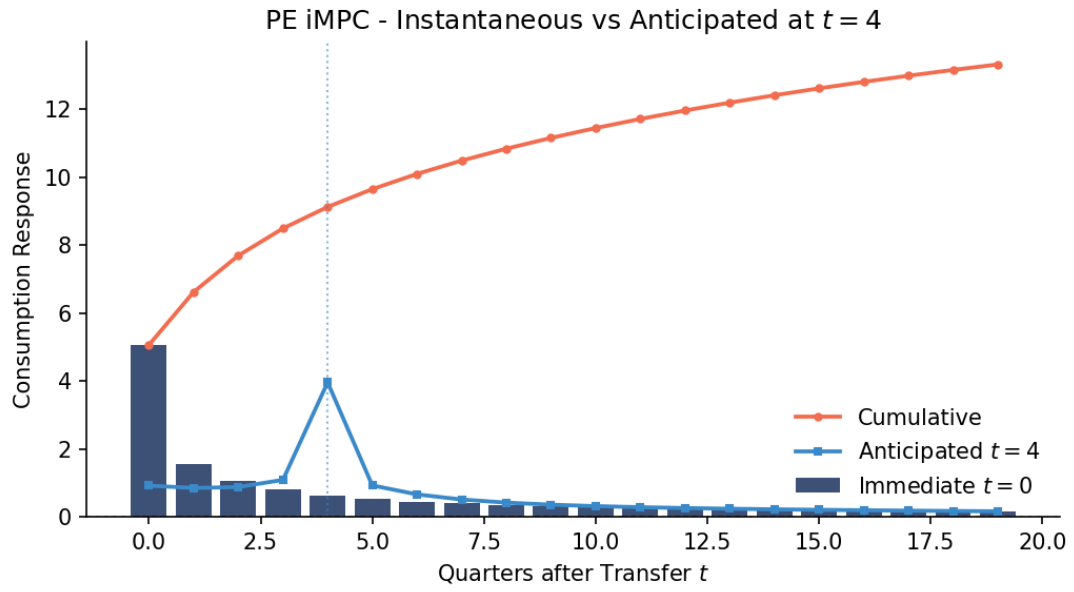


Figure 14: iMPC: Partial Equilibrium Profile

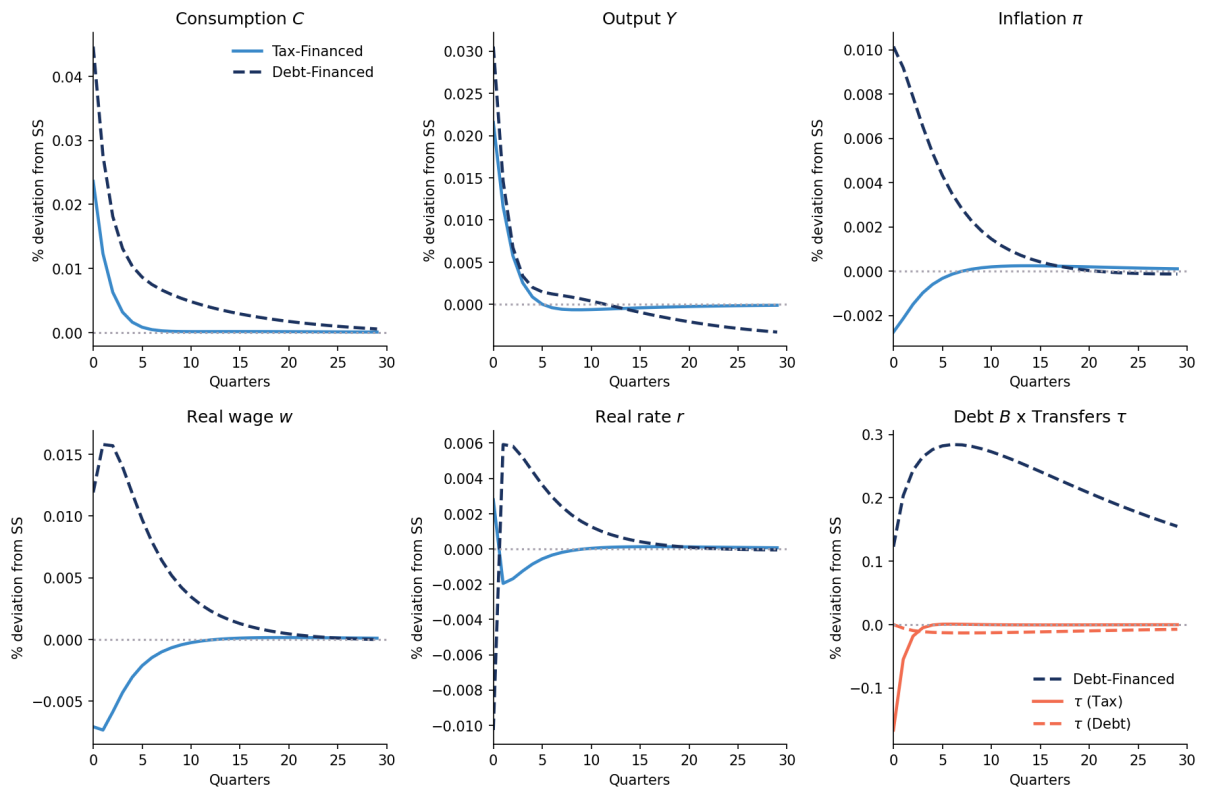


Figure 15: Response to a Shock in T by Financing Rules

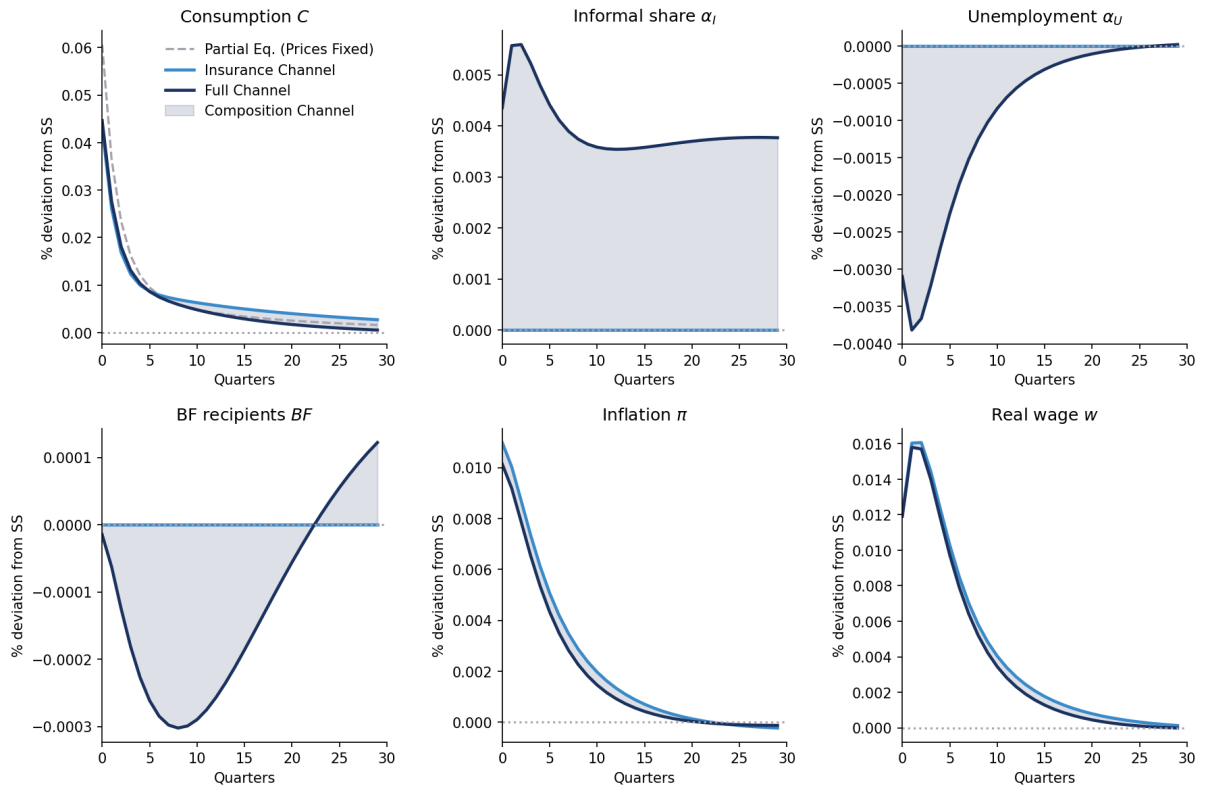


Figure 16: Response to a Shock in T by Channel

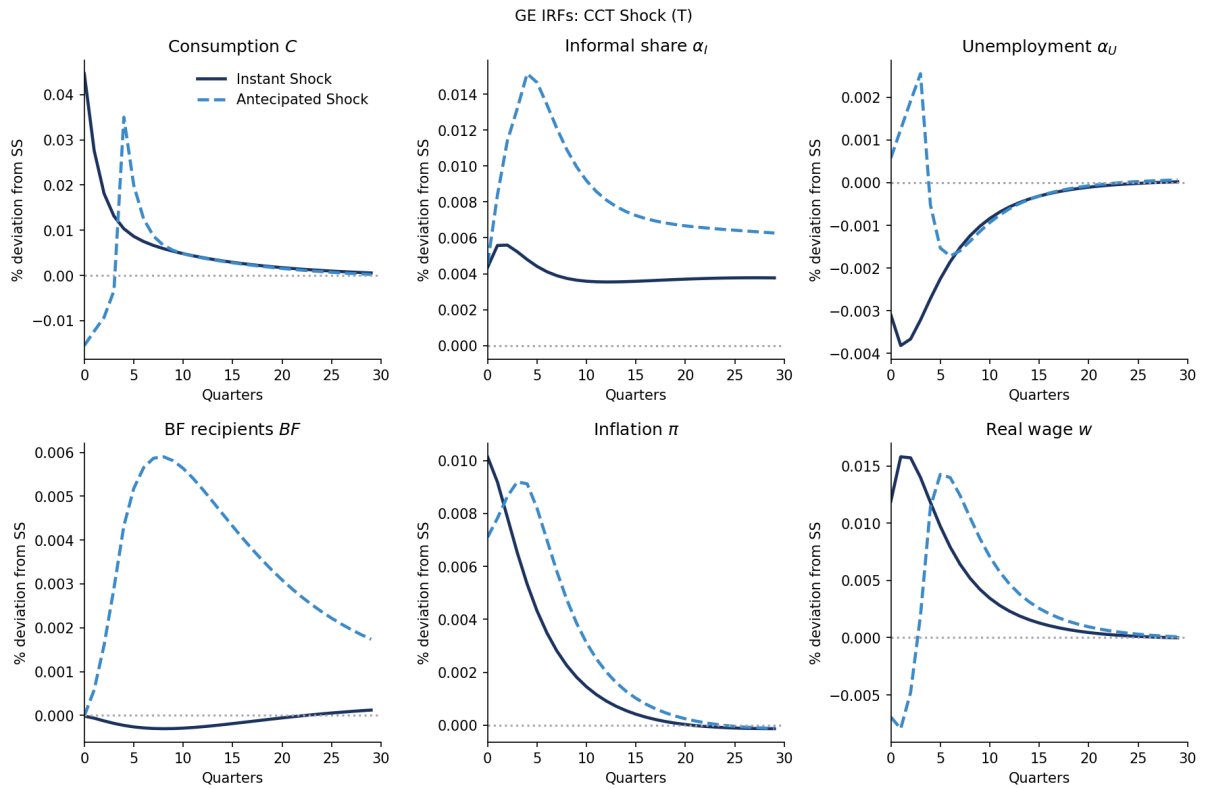


Figure 17: Response to a Shock in T : Instant vs Anticipated

Table 7: Cumulative Response (%), Shock in T

Variable	Horizon	Insurance	Full	Composition
Consumption C	1	0.043	0.045	0.001
	4	0.098	0.103	0.005
	20	0.198	0.180	-0.018
	100	0.288	0.175	-0.112
Unemployment α_U	1	0.000	-0.003	-0.003
	4	-0.000	-0.014	-0.014
	20	-0.000	-0.028	-0.028
	100	-0.000	-0.025	-0.025
Inflation π	1	0.011	0.010	-0.001
	4	0.037	0.034	-0.003
	20	0.070	0.060	-0.010
	100	0.059	0.065	0.006
Real wage w	1	0.012	0.012	-0.000
	4	0.059	0.057	-0.001
	20	0.129	0.119	-0.010
	100	0.129	0.120	-0.008

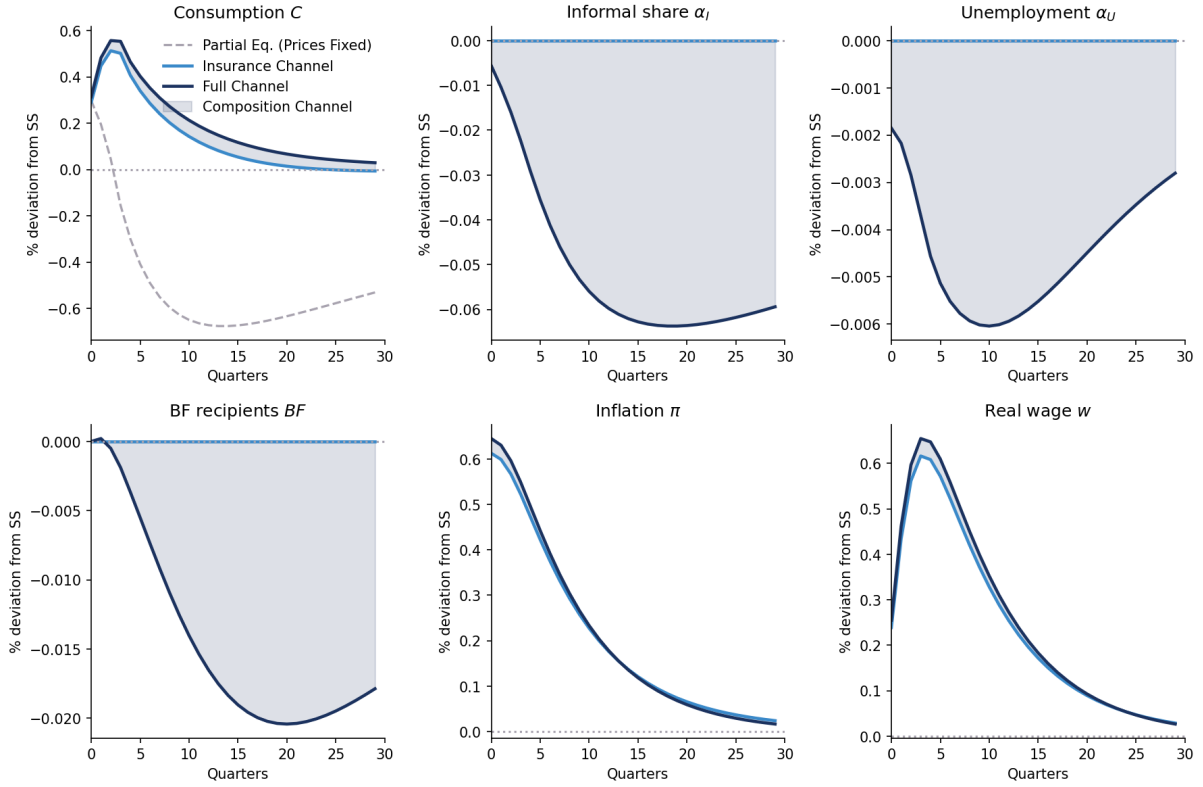


Figure 18: Response to an increase in i by Channel

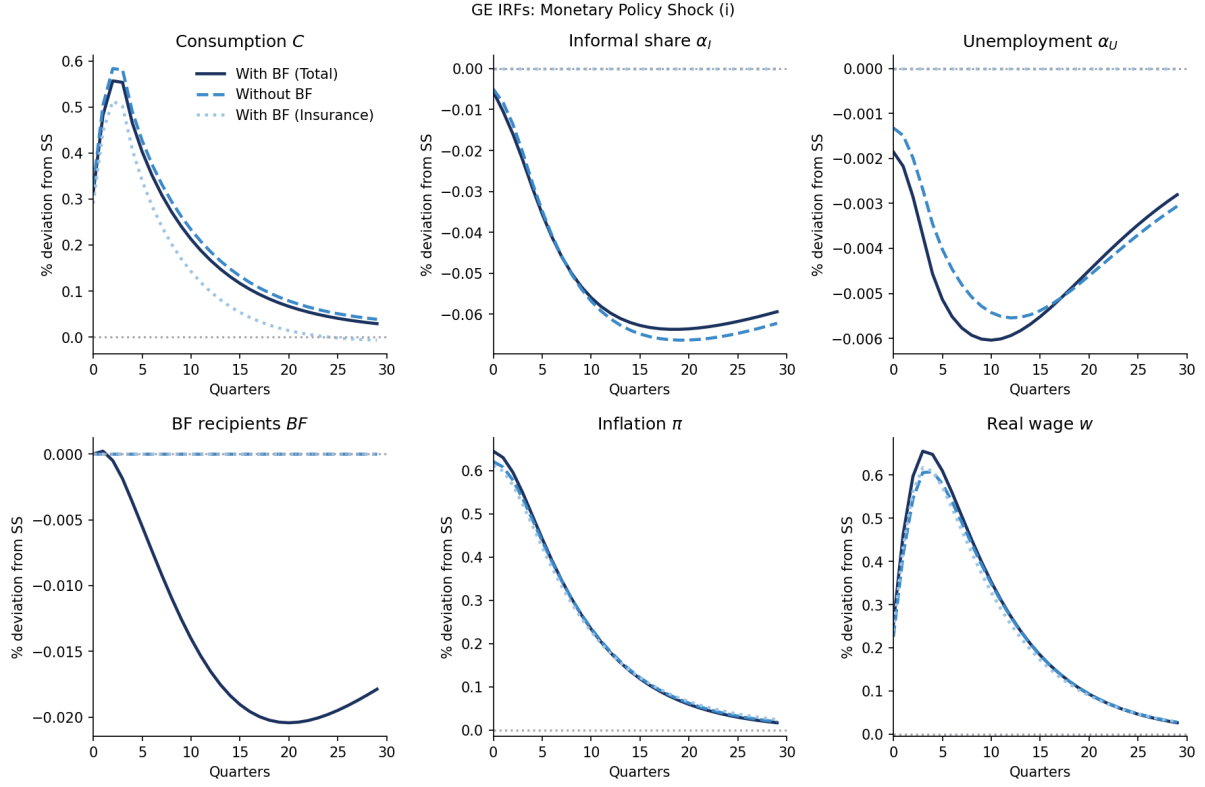


Figure 19: Response to an increase in i : With and Without CCT

Table 8: Cumulative Response (%), Shock in i

Variable	Horizon	Insurance	Full	Composition
Consumption C	1	0.290	0.314	0.024
	4	1.752	1.907	0.155
	20	4.079	5.265	1.186
	100	3.897	6.310	2.414
Unemployment α_U	1	0.000	-0.002	-0.002
	4	-0.000	-0.011	-0.011
	20	-0.000	-0.099	-0.099
	100	-0.000	-0.204	-0.204
Inflation π	1	0.613	0.645	0.032
	4	2.301	2.421	0.120
	20	5.828	6.030	0.202
	100	6.520	6.361	-0.159
Real wage w	1	0.239	0.255	0.016
	4	1.851	1.968	0.117
	20	6.777	7.230	0.453
	100	7.589	7.923	0.334

Robustness Tests

Lastly, in order to verify whether this results are representatives and robust, we intent to do some robustness tests changing some calibration parameters, such as γ , r^* (4% and 6%), and ϕ (2.0). This parameters might have minor implications in the results, but it is expected that the model continues to have the same results.

Another test would be removing self-employed and unpaid family (Conta Própria and Familiar Auxiliar) from informal households in PNAD.

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Appendix

Price Phillips Curve

Let Y_t be the aggregate output of the economy, a CES of the intermediary inputs,

$$Y_t = \left(\int_0^1 y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}}.$$

This firm maximizes its output subject to $\int_0^1 p_{i,j} y_{i,j} dj$. The FOC implies that the demand for the intermediate good is given by

$$y_{j,t} = \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t,$$

where P_t is the equilibrium price of the final good,

$$P_t = \left(\int_0^1 p_{j,t}^{1-\varepsilon} dj \right)^{\frac{1}{1-\varepsilon}}.$$

We assume a Rotemberg (1982) quadratic price adjustment cost given by

$$\frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t,$$

where $\theta_p = (\varepsilon - 1)/\kappa$ is the adjustment cost multiplier and $\bar{\Pi}$ the steady-state inflation.

Let $\Xi(y)$ be the intermediary firm's cost. Given last period's price $p_{j,t-1}$, the firm chooses this period's price $p_{j,t}$ to maximize the present discounted values of future profits,

$$V_t^{IF}(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} y_{j,t} - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}^{IF}(p_{j,t}).$$

We can replace the demand function into this problem, so

$$V_t(p_{j,t-1}) = \max_{p_{j,t}} \frac{p_{j,t}}{P_t} \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} Y_t - \Xi(y_{j,t}) - \frac{\theta_p}{2} \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right)^2 Y_t + \frac{1}{1+r_{t+1}} V_{t+1}(p_{j,t}).$$

The FOC with respect to $p_{j,t}$ is

$$(1 - \varepsilon) \left(\frac{p_{j,t}}{P_t} \right)^{-\varepsilon} \frac{Y_t}{P_t} + \varepsilon m c_{j,t} - \theta_p \left(\frac{p_{j,t}}{p_{j,t-1}} - \bar{\Pi} \right) \frac{Y_t}{p_{j,t}} + \frac{1}{1+r_{t+1}} V'_{t+1}(p_{j,t}) = 0,$$

where $m c_t = \partial \Xi / \partial y_{j,t}$, and the envelope condition gives us

$$V'_{t+1} = \theta_p \left(\frac{p_{j,t+1}}{p_{j,t}} - \bar{\Pi} \right) \frac{p_{j,t+1}}{p_{j,t}} \frac{Y_{t+1}}{p_{j,t}}.$$

Thus, taking $p_{j,t} = P_t$ in equilibrium,

$$\theta_p (\pi_t - \bar{\Pi}) \pi_t = \varepsilon \left(m c_t - \frac{\varepsilon - 1}{\varepsilon} \right) + \frac{1}{1+r_{t+1}} \theta_p (\pi_{t+1} - \bar{\Pi}) \pi_{t+1} \frac{Y_{t+1}}{Y_t},$$

where we define the firm's markup as $\mu = \frac{\varepsilon}{\varepsilon-1}$. Linearizing around the steady-state $\pi_t = \bar{\Pi}$,

$$\ln(1 + \pi_t) = \frac{\varepsilon}{\theta_p} \left(mc_t - \frac{1}{\mu} \right) + \frac{1}{1 + r_{t+1}} \ln(1 + \pi_{t+1}) \frac{Y_{t+1}}{Y_t},$$

where $\kappa = \varepsilon/\theta_p$ and $mc_t = w_t/Z_t$. This environment is equivalent to Calvo (1983) in first order around zero inflation.

Wage Phillips Curve

We will use the same setting as Hagedorn, Manovskii, and Mitman (2019), but considering just the formal sector. Assume there is a union that aggregate effective labor input into

$$h_t = \left(\int_0^1 e_{i,t} \theta_{i,t} (h_{i,t})^{\frac{\varepsilon_w-1}{\varepsilon_w}} di \right)^{\frac{\varepsilon_w}{\varepsilon_w-1}},$$

sells households labor services $e_{i,t} \theta_{i,t} h_{i,t}$ at nominal wage $W_{i,t}$ to the labor recruiter, which minimizes costs $\int_0^1 W_{i,t} e_{i,t} \theta_{i,t} h_{i,t} di$, implying a demand of

$$h_{i,t} = \left(\frac{W_{i,t}}{W_t} \right)^{-\varepsilon_w} h_t.$$

The unions sets a nominal wage $\hat{W}_t$ for an effect unit of labor, $W_{i,t} = \hat{W}_t$ to maximize profits subject to Rotemberg (1982)'s wage adjustment costs

$$e_{i,t} \theta_{i,t} \frac{\theta_w}{2} \left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w \right)^2 h_t.$$

We assume that the union's wage setting problem is to maximize the aggregate utility of its members, so

$$V_t^w = \max_{\hat{W}_t} \int \left[u'(c_{i,t}) \frac{(1 - \tau_\ell) \hat{W}_t}{P_t} e_{i,t} \theta_{i,t} h_{i,t} - v(h_{i,t}) - e_{i,t} \theta_{i,t} h_t \frac{\theta_w}{2} \left(\frac{\hat{W}_t}{\hat{W}_{t-1}} - \bar{\Pi}^w \right)^2 \right] d\Lambda_t + \bar{\beta} V_{t+1}^w.$$

Using the same algebra as the price Phillips curve, assuming that the union allocates the same amount of hours in each worker $h_{i,t} = h_t$, and using $\int_F d\Lambda_t = F_t$, the size of the formal sector,

$$\theta_w (\pi_t^w - \bar{\Pi}^w) \pi_t^w = \varepsilon_w \left[v'(h_t) - \frac{\varepsilon_w - 1}{\varepsilon_w} (1 - \tau_\ell) w_t \Lambda_t^F \right] + \bar{\beta} \theta_w (\pi_{t+1}^w - \bar{\Pi}^w) \pi_{t+1}^w \frac{h_{t+1}}{h_t},$$

where $\Lambda_t^F = \int_F e_{i,t} \theta_{i,t} u'(c_{i,t}) d\Lambda_t / F_t$. We can linearize around the steady state $\pi_t^w = \bar{\Pi}^w$ and take $h_{t+1} = h_t$, since there is no population nor technological growth. Thus,

$$\ln(1 + \pi_t^w) = \frac{\varepsilon_w}{\theta_w} \left[v'(h_t) - (1 - \tau_\ell) w_t \Lambda_t^F \frac{\varepsilon_w - 1}{\varepsilon_w} \right] + \bar{\beta} \ln(1 + \pi_{t+1}^w).$$

Therefore, defining $\kappa_w = (\varepsilon_w/\theta_w)$ at the steady state and $\mu_w = \varepsilon_w/(\varepsilon_w - 1)$, the wage

Phillips curve is

$$\ln(1 + \pi_t^w) = \kappa_w \left[\psi \cdot h_t^{1/\varphi} - \frac{(1 - \tau_\ell)w_t}{\mu_w F_t} \int_F e_{i,t} \theta_{i,t} u'(c_{i,t}) d\Lambda_t \right] + \bar{\beta} \ln(1 + \pi_{t+1}^w).$$